
Djezzy National SOC Platform

SS7 Security Module

Technical Implementation & Operations Documentation

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1. Executive Summary

The SS7 (Signaling System No. 7) Security Module represents a critical enhancement to the Djezzy National Security Operations Center (SOC) platform, providing dedicated monitoring, detection, and response capabilities for telecommunications signaling infrastructure. As mobile networks increasingly face sophisticated attacks targeting the SS7 protocol stack, this module delivers enterprise-grade protection against location tracking, fraud interception, and signaling-based denial-of-service threats.

This documentation provides comprehensive technical specifications for implementing, operating, and maintaining the complete SS7 security monitoring solution within Djezzy's on-premises infrastructure. The module integrates seamlessly with existing SOC components including Wazuh SIEM, TheHive case management, Kafka event streaming, and Elasticsearch log aggregation, creating a unified defense posture across both IP and signaling domains.

1.1 Strategic Objectives

Objective	Description	Priority
Signaling Visibility	Complete capture and normalization of SS7/MAP/ISUP/Diameter messages from STPs and HLR/VLR nodes	Critical
Attack Detection	Real-time identification of location tracking, IRSF fraud, USSD abuse, and SMS interception patterns	Critical
Regulatory Compliance	Automated ANRT reporting, audit trail maintenance, and lawful intercept support	High
Active Response	Automated blocking rules, rate limiting, and fraud team notification integration	High
Operational Efficiency	Grafana dashboards, analyst playbooks, and streamlined investigation workflows	Medium

1.2 Document Scope

This technical documentation covers all aspects of the SS7 Security Module including three core services (ss7-collector, ss7-analyzer, diameter-monitor), detection rule specifications, Kafka topic schemas, Elasticsearch index templates, Grafana dashboard configurations, deployment procedures, operations guidelines, and compliance mappings. The document is intended for security engineers, network architects, SOC analysts, and compliance officers responsible for telecommunications security at Djezzy.

Implementation Note

The SS7 module requires dedicated hardware resources (16+ vCPU, 64GB RAM, 2TB NVMe) and network mirror access from Signal Transfer Points (STPs). Ensure infrastructure prerequisites are met before beginning deployment. Refer to Chapter 3 for complete hardware specifications and network requirements.

2. Architecture Overview

The SS7 Security Module follows a microservices architecture pattern consistent with the broader Djezzy SOC platform design philosophy. Three specialized services handle distinct aspects of the signaling security pipeline: message collection, threat analysis, and Diameter/LTE interface monitoring. Each service operates as an independent Docker container with well-defined interfaces, enabling horizontal scaling and fault isolation.

2.1 System Architecture

Component	Technology Stack	Role	Network
ss7-collector	Python 3.11 + Kamailio + libss7	SIGTRAN/M3UA capture, message normalization	soc-events
ss7-analyzer	Python 3.11 + Scikit-learn	Rule engine, ML scoring, alert generation	soc-backend/soc-events
diameter-monitor	Python 3.11 + pyDiameter	LTE/EPS S6a/S6d/Gx/Rx/Cx decoding	soc-events
Kafka Cluster	Apache Kafka 3.x	Event streaming, buffer layer	soc-events
Elasticsearch	OpenSearch 2.x	Indexing, search, retention	soc-backend
TheHive	TheHive 5.x	Case management, investigation tracking	soc-backend
Grafana	Grafana 10.x	Visualization, dashboards, alerting	soc-monitoring

2.2 Data Flow Architecture

The data flow through the SS7 module follows a structured pipeline designed for high throughput (target: 50K events per second) and low-latency detection (sub-second alerting). Raw signaling messages enter through the ss7-collector service via port mirroring from STPs, undergo protocol parsing and normalization, then stream into Kafka topics for distributed consumption by downstream analysis services.

Stage	Input	Processing	Output	Latency
1. Capture	SCTP/M3UA packets	Protocol decode, PCAP write	Normalized JSON	<10ms
2. Buffer	JSON messages	Kafka partitioning, replication	Kafka topics	<50ms
3. Analyze	Kafka streams	Rule matching, ML scoring	Alerts, scores	<500ms
4. Store	Alerts, raw events	Elasticsearch indexing	Searchable indices	<2s
5. Respond	High-severity alerts	Case creation, blocking	TheHive, firewall	<5s

2.3 Network Integration Points

The SS7 module integrates with Djezzy's core network at multiple points to achieve comprehensive visibility. Primary data sources include Signal Transfer Points (STPs) for inter-network signaling, Home Location Registers (HLR) for subscriber database queries, Visitor Location Registers (VLR) for roaming subscriber tracking, and Mobile Switching Centers (MSC) for call setup telemetry. Network taps or SPAN ports provide passive monitoring capability without impacting production traffic.

Source Node	Protocol	Port	Data Type	Volume Estimate
STP Primary	M3UA/SCTP	2904/2905	Inter-network MAP/ISUP	15K msg/s
STP Secondary	M3UA/SCTP	2904/2905	Redundant path messages	15K msg/s
HLR	MAP	Internal	Subscriber queries, updates	8K msg/s
VLR	MAP	Internal	Location updates, roaming	5K msg/s
MSC	ISUP/TUP	Internal	Call setup, teardown	7K msg/s
HSS (LTE)	Diameter	3868	S6a authentication	3K msg/s
PCRF	Diameter	3868	Gx/Rx policy	2K msg/s

Network Security Requirement

All SS7 monitoring interfaces **MUST** be deployed on isolated VLANs with strict access controls. The collector service should receive only mirrored traffic (RX-only) to prevent any possibility of injected packets affecting production signaling. Refer to the network policies in Appendix A for detailed ACL configurations.

3. Service Specifications

This chapter provides detailed technical specifications for each service in the SS7 Security Module. Each service is designed as a containerized microservice with explicit configuration parameters, resource allocations, and operational characteristics. Services communicate through well-defined interfaces using Apache Kafka for asynchronous messaging and REST APIs for synchronous management operations.

3.1 SS7 Collector Service (ss7-collector)

The ss7-collector service serves as the primary ingestion point for all SS7 and SIGTRAN traffic. Built on Python 3.11 with Kamailio integration for SIP/SS7 translation and libss7 for low-level MTP3 processing, this service captures signaling packets from mirrored network interfaces, decodes protocol layers (MTP2, MTP3, SCCP, TCAP, MAP, ISUP, CAP), normalizes message structures to JSON format, and publishes to Kafka topics for downstream analysis.

3.1.1 Container Configuration

Parameter	Value	Description
Image	djezzy-soc/ss7-collector:latest	Custom-built image based on python:3.11-slim
CPU Limit	8.0 cores	Dedicated processing for packet decoding
Memory Limit	16 GB	Message buffering, state tables, PCAP buffers
Storage Mount	/var/capture/ss7 (2TB NVMe)	PCAP retention, rotated daily
Config Volume	/etc/ss7-collector	Rule files, GT translations, whitelists
Listen Ports	2904 (M3UA), 2905 (SCTP), 3868 (Diameter)	Passive monitoring only
Network	soc-events (isolated)	No external connectivity required
Health Check	:7000/health	HTTP endpoint, 30s interval, 5s timeout

3.1.2 Environment Variables

Variable	Required	Default	Description
SS7_LISTEN_INTERFACE	Yes	eth0	Network interface for packet capture
SS7_M3UA_PORT	Yes	2904	M3UA (MTP3 User Adaptation) port
SS7_SCTP_PORT	Yes	2905	SCTP transport layer port
DIAMETER_PORT	No	3868	Diameter protocol listener port
KAFKA_BOOTSTRAP_SERVERS	Yes	-	Comma-separated broker list
OUTPUT_TOPIC	Yes	ss7-raw-events	Primary Kafka output topic

Variable	Required	Default	Description
PCAP_RETENTION_DAYS	No	30	Local PCAP file retention period
MESSAGE_BUFFER_SIZE	No	100000	In-memory message buffer capacity
DEBUG_LOG_LEVEL	No	INFO	Logging verbosity (DEBUG/INFO/WARN/ERROR)

3.2 SS7 Analyzer Service (ss7-analyzer)

The ss7-analyzer service performs real-time threat detection on normalized SS7 messages consumed from Kafka. It implements a multi-layer detection engine combining signature-based rule matching (YAML-configurable), behavioral anomaly detection (Scikit-learn models), and correlation analysis across message streams. Detected threats generate alerts with risk scores, MITRE ATT&CK; technique classifications, and automated response recommendations.

3.2.1 Detection Engine Architecture

Layer	Technique	Detection Capability	Latency
Layer 1 - Signature	Pattern matching (YAML rules)	Known attack signatures, fraud patterns	<10ms/msg
Layer 2 - Behavioral	Statistical profiling, ML models	Anomalous query volumes, unusual patterns	<50ms/window
Layer 3 - Correlation	Temporal/spatial correlation	Multi-stage attack sequences	<500ms/correlation
Layer 4 - Contextual	Threat intelligence enrichment	IOC matching, reputation scoring	<2s/enrichment

3.2.2 Alert Classification Schema

Severity	Criteria	Auto-Response	Notification
CRITICAL	Confirmed attack pattern, financial impact >\$10K/hour	Block source GT, create P1 case	Fraud team + SOC manager + on-call
HIGH	Strong indicator, potential privacy violation	Rate limit source, create P2 case	SOC analysts + shift lead
MEDIUM	Suspicious pattern, requires investigation	Log for forensics, flag analyst	SOC analyst queue
LOW	Informational anomaly, baseline deviation	Metric collection only	Dashboard update only

3.3 Diameter Monitor Service (diameter-monitor)

The diameter-monitor service extends SS7 security coverage to LTE/EPS (Evolved Packet System) signaling interfaces. As Djezzy deploys 4G/LTE infrastructure, Diameter-based protocols replace traditional SS7 for authentication (S6a), subscriber data (S6d), policy control (Gx/Rx), and interconnect (Cx). This service decodes AVP (Attribute-Value Pair) structures, monitors for LTE-specific attacks including IMSI catchers, fake base station detection, and subscriber data exfiltration attempts.

Interface	Application	Monitored Operations	Attack Vectors
S6a	Home Subscriber Server (HSS)	Authentication, location update, subscriber MSA	MSA catcher, tracking, data theft
S6d	Subscription Data Repository	Profile retrieval, service authorization	Unauthorized profile access
Gx	Policy and Charging Rules (PCRF)	QoS policy installation, charging rules	Policy injection, free data fraud
Rx	Application Function (AF)	Session binding, media authorization	Session hijacking, bypass
Cx	Call Session Control (IMS)	User registration, route lookup	Registration hijacking, intercept

Scaling Recommendation

For production deployments handling peak traffic (>40K EPS), deploy multiple instances of each service behind Kafka consumer groups. The ss7-analyzer benefits most from horizontal scaling as CPU-bound rule evaluation can be partitioned by message type or source GT range. Target 3-5 analyzer instances for full coverage.

4. Detection Rules Reference

The SS7 Security Module implements a comprehensive rule set designed to detect known attack patterns specific to telecommunications signaling infrastructure. Rules are defined in YAML format for human readability and rapid modification, then compiled to optimized internal structures at service startup. This chapter catalogs all production rules organized by attack category, with technical details on detection logic, threshold configuration, and response actions.

4.1 Location Tracking Detection

Location tracking attacks exploit the Send Routing Info For SM (SRI4SM) MAP operation to determine a subscriber's current serving MSC/VLR address without authorization. Attackers may be stalkers, private investigators, or state actors targeting individuals of interest. The detection rule identifies suspicious query patterns including excessive queries to a single MSISDN, queries from untrusted Global Titles, and timing correlations suggesting tracking behavior rather than legitimate SMS delivery attempts.

Parameter	Configuration	Rationale
Rule ID	SS7_LOCATION_TRACKING_SUSPICIOUS	Unique identifier for alert correlation
Message Type	MAP_SEND_ROUTING_INFO_FOR_SM	Primary target operation
Threshold Count	10 requests	Per-subscriber, per 60-second window
Aggregation Key	source_global_title + imsi_prefix	Group by originator
Severity	HIGH	Privacy violation risk
MITRE Technique	T1419 (SIM Card Swap)	ATT&CK framework mapping
Action 1	create_thehive_case	Automatic incident creation
Action 2	send_alert_wazuh	SIEM notification
Action 3	block_source_gt_if_repeated	Conditional blocking (>50/hour)

4.2 International Revenue Share Fraud (IRSF)

IRSF represents one of the most financially damaging telecom fraud types, costing the global industry billions annually. Attackers compromise PBX systems or exploit weak SIP passwords to generate high-volume international calls to premium-rate numbers they control, sharing revenue with complicit destination carriers. The SS7 module detects IRSF by analyzing ISUP Initial Address Messages (IAM) for high-risk destination patterns, short-duration calls (characteristic of revenue-generation rather than conversation), and abnormal calling patterns.

Parameter	Configuration	Rationale
Rule ID	SS7_IRSF_PATTERN_DETECTED	Fraud-specific identifier
Message Type	ISUP_IAM (Initial Address Message)	Call setup initiation
Destination Filter	high_risk_countries list	Configurable country blacklist
Duration Pattern	short_calls (<10 seconds)	Revenue generation signature
Calling Number Type	premium_rate detection	Source number analysis
Volume Threshold	100 calls / 300 seconds	Bulk fraud detection
Severity	CRITICAL	Financial impact classification
Actions	P1 case, fraud notify, block, CDR report	Full response chain

4.3 USSD Brute Force Detection

Unstructured Supplementary Service Data (USSD) sessions enable interactive services such as balance inquiries, mobile money transfers, and account management. Attackers may brute-force USSD codes to enumerate valid accounts, probe service availability, or trigger unintended operations. The detection rule monitors for excessive USSD request rates per MSISDN, focusing on sensitive service codes (*100# balance, *101# transfer, *123# services) that could indicate account enumeration or service abuse attempts.

Parameter	Configuration	Notes
Rule ID	SS7_USSD_BRUTE_FORCE	USSD-specific detection
Message Type	MAP_PROCESS_UNSTRUCTURED_SS_REQUEST	USSD invocation
Monitored Codes	*100#, *101#, *123#, *124#, *150#	Sensitive service codes
Threshold	30 requests / 60 seconds	Per-MSISDN rate limit
Severity	MEDIUM	Service abuse, not critical infrastructure
Actions	rate_limit_msisdn, alert_analyst, log_forensics	Progressive response

4.4 SMS Interception Indicators

SMS interception attacks forward victim messages to attacker-controlled destinations, enabling content theft (OTP codes, banking alerts) and surveillance. Detection focuses on the MAP Forward Short Message operation, identifying suspicious forwarding patterns including multiple distinct destinations, international high-risk countries, business-hours-only activity (suggesting automated systems), and content matching sensitive patterns like OTP formats or banking keywords.

Indicator	Threshold	Detection Logic
Multiple Destinations	>5 unique in 300s	Distribution attack pattern
International High-Risk	Country code match	Geographic risk scoring
Timing Anomaly	Business hours only	Automated system indicator
Content Match	OTP/banking regex	Payload inspection (privacy note)
Severity	HIGH	Subscriber privacy violation
MITRE Technique	T1421 (SMS Interception)	Framework alignment

Privacy Consideration for Content Inspection

SMS content inspection for OTP/banking pattern matching requires careful legal review. In Algeria, ANRT regulations permit fraud-related content analysis but mandate encryption at rest and strict access logging. Implement content scanning only after obtaining proper legal authorization and ensure all inspected payloads are encrypted in Elasticsearch.

5. Integration Guide

The SS7 Security Module integrates with multiple components of the Djazzy SOC platform to provide end-to-end security monitoring capabilities. This chapter details the technical integration points including Kafka topic schemas for event streaming, Elasticsearch index templates for log storage and search, Grafana dashboard configurations for visualization, and TheHive API integrations for automated case management.

5.1 Kafka Topic Configuration

Apache Kafka serves as the central event streaming backbone for the SS7 module, providing durable buffering, horizontal scalability, and replay capability for debugging. Four dedicated topics handle different message types with partitioning strategies optimized for throughput and consumer parallelism. All topics use replication factor 3 for fault tolerance across the three-broker Kafka cluster.

Topic Name	Partitions	Replication	Retention	Purpose
ss7-raw-events	12	3	7 days	Normalized SS7/Diameter messages from collector
ss7-alerts	6	3	30 days	Detected threats with scores and context
diameter-events	6	3	7 days	Decoded LTE/EPS Diameter AVPs
ss7-fraud-indicators	4	3	90 days	Aggregated fraud metrics for reporting

5.2 Elasticsearch Index Template

Elasticsearch (deployed as OpenSearch 2.x for on-premises compatibility) stores indexed SS7 events for search, analytics, and compliance retention. The index template defines field mappings optimized for signaling data types including keyword fields for exact-match lookups (GT, MSISDN, IMSI prefix), date fields for time-range queries, geo_point fields for location analysis, and float fields for machine learning risk scores.

Field Name	Type	Description	Use Case
timestamp	date	Message reception time	Time-range queries, retention
message_type	keyword	MAP/ISUP/Diameter operation	Filtering by protocol operation
calling_party	keyword	Originating address (GT/MSISDN)	Source tracking, blacklists
called_party	keyword	Destination address	Destination analysis
imsi	keyword + text	Subscriber identity (partial match)	Subscriber correlation
global_title	keyword	SS7 signaling point address	Network topology mapping
risk_score	float	ML-generated threat score (0-100)	Prioritization, thresholding

Field Name	Type	Description	Use Case
attack_category	keyword	Classification (fraud/tracking/DoS)	Dashboard filtering
location_info	geo_point	VLR/MSC geographic coordinates	Geospatial visualization
roaming_status	keyword	Home/Visitor/International Roaming	Roaming fraud detection

5.3 Grafana Dashboard Configuration

Two primary Grafana dashboards provide real-time visibility into SS7 security posture. The SS7 Security Overview dashboard offers operational metrics including message volume by type, top alert sources, geographic distribution of threats, and active investigation counts. The IRSF Fraud Detection dashboard focuses specifically on revenue protection with high-risk destination rankings, call duration analysis, revenue-at-risk calculations, and blocking rule effectiveness metrics.

Dashboard	Panel	Data Source	Visualization
SS7 Security Overview	Messages/sec by Type	Elasticsearch (ss7-*)	Time series bar
SS7 Security Overview	Top 10 Alert Sources (GT)	Elasticsearch (ss7-alerts)	Ranking table
SS7 Security Overview	Geographic Threat Heatmap	Elasticsearch (geo_point)	World map
SS7 Security Overview	Active TheHive Cases	TheHive API	Stat panel
SS7 Security Overview	24h Alert Timeline	Elasticsearch (ss7-alerts)	Timeline
IRSF Fraud Detection	High-Risk Destinations	Elasticsearch (ss7-fraud)	Ranking table
IRSF Fraud Detection	Call Duration Distribution	Elasticsearch (ss7-raw)	Histogram
IRSF Fraud Detection	Revenue at Risk (\$)	Prometheus (custom)	Gauge + trend
IRSF Fraud Detection	Blocking Rule Effectiveness	Elasticsearch (ss7-alerts)	Stat + sparkline

5.4 TheHive Case Management Integration

The ss7-analyzer service automatically creates TheHive cases for HIGH and CRITICAL severity alerts, enabling structured investigation workflows, evidence attachment, and status tracking. Integration uses TheHive's REST API v1 with API key authentication, mapping SS7 alert fields to case custom fields for telecom-specific investigation context.

SS7 Alert Field	TheHive Case Field	Mapping Logic
alert_id	case.tags[]	Tag: ss7-{rule_id}
severity	case.severity	CRITICAL->2, HIGH->1, MEDIUM->0.5, LOW->0.2
attack_category	case.title	Prefix: [{category}] {summary}

SS7 Alert Field	TheHive Case Field	Mapping Logic
source_global_title	case.customFields.source_gt	String field
target_msisdn	case.customFields.target_subscriber	String field
risk_score	case.customFields.ml_score	Float field (0-100)
raw_message	case.observable[]	Type: other, value: JSON payload
detection_rules[]	case.description	Formatted rule list

API Rate Limiting

TheHive enforces API rate limits (default: 30 requests/second). During high-volume alert periods (e.g., detected attack campaigns), implement client-side batching with 5-second intervals between case creations. Queue overflow should log to Kafka dead-letter topic for later manual review rather than dropping alerts silently.

6. Deployment Procedures

Deploying the SS7 Security Module requires careful orchestration across network infrastructure, container orchestration, and security tooling integration. This chapter provides step-by-step procedures organized into three phases: passive monitoring setup (weeks 1-2), rule activation (weeks 3-4), and active response enablement (weeks 5-6). The phased approach allows baseline calibration, false-positive tuning, and gradual trust-building before automated actions affect production signaling.

6.1 Phase 1: Passive Monitoring Setup (Weeks 1-2)

6.1.1 Network Tap Configuration

Before deploying the collector service, establish network visibility to SS7 traffic flows. Coordinate with network engineering to configure SPAN ports on core switches connected to STPs, or deploy optical taps on fiber links carrying signaling traffic. Verify that mirrored traffic includes both M3UA (port 2904) and SCTP (port 2905) protocols with sufficient bandwidth headroom (recommend 2x expected peak traffic volume).

```
# Configure SPAN session on Cisco Catalyst switch (example)
configure terminal
monitor session 1 source interface TenGigabitEthernet1/1 both
monitor session 1 destination interface TenGigabitEthernet1/2 encapsulation replicate
no monitor session 1 filter packet-type good
exit
show monitor session 1

# Verify traffic capture on collector server
tcpdump -i eth0 port 2904 or port 2905 -c 100
```

6.1.2 Docker Compose Deployment

Deploy the SS7 services using Docker Compose with the production overlay file. The compose file defines resource limits, network attachments, volume mounts, and health checks for each service. Start with the collector service first to validate message capture before enabling downstream analyzers.

Step	Command	Validation
1. Pull images	docker compose -f docker-compose.prod.yml pull ss7-collector ss7-analyzer ss7-diameter-monitor	Verify image digests
2. Create volumes	docker volume create ss7-pcaps && docker volume create ss7-rules	List volumes
3. Start collector	docker compose -f docker-compose.prod.yml up -d ss7-collector	Check logs, health endpoint
4. Validate capture	kafka-console-consumer --topic ss7-raw-events --max-messages 50	Confirm JSON messages
5. Start analyzer	docker compose -f docker-compose.prod.yml up -d ss7-analyzer	Check rule loading logs
6. Start diameter	docker compose -f docker-compose.prod.yml up -d diameter-monitor	Verify Diameter decode
7. Full pipeline test	Generate test SS7 message, trace through pipeline	End-to-end latency <5s

6.2 Phase 2: Rule Activation (Weeks 3-4)

With passive capture validated and baseline traffic profiles established, begin activating detection rules in read-only mode (alerts only, no blocking actions). This phase focuses on threshold calibration, false-positive reduction, and analyst workflow integration. Start with lower-severity rules to generate manageable alert volumes while tuning detection sensitivity.

Task	Owner	Deliverable	Sign-off
Enable LOW severity rules	SOC Engineer	Baseline metric comparison	Team Lead
Review alert samples (Day 1-3)	SOC Analysts	False positive log	Security Architect
Adjust thresholds based on FP rate	SOC Engineer	Updated rule configs	Team Lead
Enable MEDIUM severity rules	SOC Engineer	Alert volume metrics	Manager
Configure TheHive case templates	TheHive Admin	Custom field definitions	SOC Lead
Train analysts on SS7 investigation	Security Trainer	Training completion records	HR/Manager
Enable HIGH severity rules (read-only)	SOC Engineer	Alert quality assessment	Architect

6.3 Phase 3: Active Response (Weeks 5-6)

The final phase enables automated response actions for confirmed attack patterns. Active response includes automatic TheHive case creation, source GT rate-limiting recommendations pushed to firewall policy, fraud team notifications via PagerDuty/Slack, and regulatory report generation for ANRT compliance. Enable responses incrementally, starting with notification-only actions before progressing to blocking capabilities.

Change Management Requirement

Active response activation requires formal change approval from Djezzy's Change Advisory Board (CAB) due to potential impact on production signaling. Prepare rollback procedures documenting how to immediately disable all automated actions and revert to monitoring-only mode. Test rollback during a scheduled maintenance window before go-live.

7. Operations & Maintenance

Ongoing operations of the SS7 Security Module follow established SOC procedures with telecommunications-specific adaptations. This chapter documents daily operational tasks, monitoring dashboards, backup procedures, incident response playbooks for SS7-specific threats, and routine maintenance activities including rule updates and model retraining.

7.1 Daily Operational Procedures

Time	Task	Frequency	Owner	Tools
00:00	Backup PCAP rotations to cold storage	Daily	NOC	Scripts + rsync
06:00	Review overnight alert summary	Daily	Morning Shift	Grafana + TheHive
09:00	Check Kafka consumer lag	Daily	SOC Engineer	Kafka UI / CLI
10:00	Validate Elasticsearch index health	Daily	SOC Engineer	OpenSearch Dashboard
12:00	Process pending TheHive cases	As-needed	SOC Analysts	TheHive Web UI
14:00	Update threat intel feeds (GT blacklists)	Daily	Intel Team	MISP/OpenCTI
16:00	Generate daily metrics report	Daily	SOC Lead	Grafana + Scripts
18:00	Shift handover briefing	Daily	Shift Leads	Documentation

7.2 Health Monitoring & Alerting

Prometheus metrics exported by each SS7 service enable proactive monitoring of system health and performance degradation. Critical alerts should route to the NOC on-call rotation via PagerDuty or equivalent escalation system. Define alert thresholds based on baseline metrics collected during Phase 1 passive monitoring.

Alert Name	Expression	Threshold	Severity	Response
Ss7CollectorDown	up{job="ss7-collector"} == 0	Any	CRITICAL	Immediate restart
Ss7CollectorHighLatency	histogram_quantile(0.99, ss7_process_seconds) > 1>1s		WARNING	Investigate load
KafkaConsumerLag	kafka_consumer_group_lag > 10000	>10K	WARNING	Scale consumers
ElasticsearchIndexErrors	rate(es_index_errors_total[5m]) > 0.1	>0.1/s	WARNING	Check cluster
Ss7AnalyzerErrorRate	rate(ss7_analyze_errors_total[5m]) > 0.05	>5%	CRITICAL	Check rules
DiskSpaceLow	ss7_pcap_disk_usage_percent > 85	>85%	WARNING	Expand storage
AlertBacklog	ss7_alert_queue_length > 1000	>1K	WARNING	Scale analyzer

7.3 Backup & Retention Procedures

SS7 data requires extended retention periods for forensic investigation and regulatory compliance. Implement a tiered storage strategy balancing cost against accessibility: hot storage (NVMe SSD) for 7 days of searchable indices, warm storage (HDD array) for 90 days of compressed archives, and cold storage (tape/object storage) for 7-year regulatory retention. PCAP files capturing raw signaling packets require the longest retention as they constitute immutable evidence.

Tier	Storage	Retention	Data Types	Access Time
Hot	NVMe SSD (local)	7 days	Elasticsearch indices, recent PCAPs	<1 second
Warm	HDD NAS (network)	90 days	Compressed PCAPs, aggregated metrics	<5 minutes
Cold	Object storage (S3-compatible)	7 years	Encrypted archives, monthly summaries	<4 hours
Archive	WORM tape (offsite)	Permanent	Incident-related exports, ANRT reports	<48 hours

7.4 SS7-Specific Incident Response Playbooks

SS7 incidents require specialized investigation procedures distinct from traditional IT security events. Analysts must understand signaling protocol semantics, network topology implications, and regulatory notification requirements. The following playbooks provide structured response guidance for common SS7 threat scenarios encountered in telecommunications environments.

Playbook	Trigger	Initial Response	Escalation	Resolution Criteria
Location Tracking	SRI4SM threshold breach	Identify target/victim, block source	Privacy officer (subscriber)	Source blocked, victim notified
IRSF Campaign	IAM volume spike to premium destinations	Block destination ranges, isolate fraud	Fraud team, legal	Revenue leak stopped, CDRs preserved
USSD Abuse	Brute force threshold hit	Rate limit MSISDN, identify intent	Customer care (if subscriber)	Account secured, pattern documented
SMS Intercept	Forwarding pattern detected	Preserve evidence, identify recipient	Law enforcement liaison	Attack stopped, legal process initiated
Signaling DoS	Message flood from single GT	Traffic filtering at STP	Network ops, vendor support	Normal traffic restored, root cause fixed

Evidence Preservation

For any incident potentially involving criminal activity (fraud, interception, stalking), immediately preserve raw PCAP files and export relevant Elasticsearch documents with hash integrity verification. Chain of custody documentation must begin at initial detection, not when law enforcement involvement becomes likely. Consult legal counsel before destroying any SS7 data regardless of standard retention policies.

8. Regulatory Compliance

Operating as a licensed telecommunications carrier in Algeria, Djezzy must comply with ANRT (Autorite de Regulation de la Poste et des Communications Electroniques) regulations governing network security, subscriber privacy, and fraud prevention. The SS7 Security Module directly supports multiple compliance requirements through its monitoring, detection, and reporting capabilities. This chapter maps module features to specific regulatory obligations and documents audit evidence artifacts.

8.1 ANRT Compliance Matrix

Requirement	ANRT Reference	Module Feature	Evidence Artifact	Audit Frequency
Signaling attack logging	Art. 42-44 Cybersecurity Law	Full MAP/ISUP/Diameter capture	Elasticsearch indices, 90-day retention	Quarterly
Incident reporting	Circular 05/2023	Automated ANRT portal reports	Monthly summary + immediate alerts	Monthly + ad-hoc
Subscriber privacy protection	Data Protection Law Art. 8	Location query monitoring	TheHive cases per suspicious IP	Annual
Fraud prevention framework	Telecom Regulations Art. 38	IRSF detection & blocking	Dashboard metrics, blocked GT	Quarterly
Lawful intercept support	CI Act implementation	LI interface logging	Separate audit trail, encrypted	Annual (classified)
Staff training records	Cybersecurity directive 03/2022	SIEM analyst certification tracking	Training LMS completion records	Annual
Vulnerability management	ANRT security guidelines	SS7 vulnerability scanning	Scan reports, remediation tickets	Monthly
Business continuity	Telecom resilience standards	DR site failover capability	Failover test results	Bi-annual

8.2 Audit Evidence Generation

Regulatory audits require documentary evidence demonstrating compliance controls are operating effectively. The SS7 module automates evidence collection through scheduled report generation, tamper-evident logging, and cryptographic integrity verification. Maintain an evidence repository with version-controlled documents accessible to authorized auditors while protecting sensitive subscriber data from unauthorized exposure.

Evidence Type	Generation Method	Format	Retention	Access Control
Message capture logs	Continuous (collector service)	PCAP + JSON metadata	7 years	SOC + Legal only
Alert disposition records	On-demand + scheduled	PDF + CSV exports	7 years	Compliance team
Rule effectiveness metrics	Weekly automated script	Grafana dashboard snapshots	3 years	SOC management
Incident response timelines	TheHive case export	JSON + PDF case files	7 years	Legal + SOC
System configuration audits	Monthly diff reports	Git-managed config files	3 years	Security architect
Access logs (who viewed what)	Elasticsearch audit index	Structured JSON logs	3 years	Compliance only

Evidence Type	Generation Method	Format	Retention	Access Control
Training completion records	LMS integration	Certificates + transcripts	Employee tenure	HR + Compliance

8.3 Regulatory Reporting Automation

The SS7 module includes automated report generation for ANRT submission requirements. Monthly security summaries aggregate alert statistics, incident counts, and trend analysis into standardized formats compatible with the ANRT online reporting portal API. Critical security incidents trigger immediate notification workflows with 72-hour submission deadlines as mandated by cybersecurity regulations. Report templates are version-controlled and reviewed quarterly for regulatory changes.

Report	Frequency	Deadline	Contents	Recipient
Monthly Security Summary	Monthly	5th of following month	Alert stats, incident count, trends	ANRT Portal API
Critical Incident Notification	Ad-hoc	Within 72 hours	Impact assessment, response actions	ANRT + Internal Mgmt
Quarterly Fraud Metrics	Quarterly	15th of quarter-end month	IRSF attempts blocked, revenue saved	ANRT + Finance
Annual Compliance Attestation	Annually	January 31	Control effectiveness evidence	ANRT + Board
Lawful Intercept Statistics	Semi-annually	June 30 / Dec 31	Request counts, response times (classified)	ANRT (secure channel)

Classified Information Handling

Lawful Intercept statistics and certain fraud investigation details contain classified information subject to Algerian national security regulations. Store these materials in separate, encrypted repositories with access restricted to cleared personnel. Never include classified content in regular audit packages shared with external auditors without proper security clearance verification and non-disclosure agreements.

9. Troubleshooting Guide

This chapter addresses common operational issues encountered during SS7 module deployment and ongoing operations. Each troubleshooting entry includes symptom description, diagnostic steps, root cause analysis, and resolution procedures. Issues are categorized by affected component to facilitate rapid lookup during incident response scenarios.

9.1 SS7 Collector Issues

Symptom	Possible Cause	Diagnostic Command	Resolution
No messages captured	SPAN misconfiguration, wrong interface	<code>tcpdump -i eth0 port 2904</code>	Verify SPAN config, check interface
High packet drop rate	Insufficient CPU, NIC ring buffer too small	<code>cat /proc/net/dev; top -H</code>	Increase CPU limit, tune NIC buffers
Kafka publish failures	Broker unreachable, topic not created	<code>terraform-console-producer --test</code>	Check broker health, create topics
PCAP rotation errors	Disk full, permission issues	<code>df -h; ls -la /var/capture/</code>	Expand storage, fix permissions
M3UA decode errors	Protocol version mismatch, corrupt packets	Check logs for decode exceptions	Update decoder, verify mirror integrity

9.2 SS7 Analyzer Issues

Symptom	Possible Cause	Diagnostic Command	Resolution
High alert latency	Consumer lag, overloaded rules engine	<code>kafka-consumer-groups --describe</code>	Add consumer instances, optimize rules
Excessive false positives	Thresholds too sensitive, baseline queries	Query ES for alert/dismissal ratio	Recalibrate thresholds, update baselines
Rules not loading	YAML syntax error, missing dependencies	Check startup logs for parse errors	Fix YAML syntax, validate rule deps
ML model scoring errors	Model file corrupted, version mismatch	Check model checksum, test inference	Retrain/redeploy model
TheHive API errors	Rate limit exceeded, auth token expired	<code>curl -H "Authorization: {key}" thehiveapi.com</code>	Implement backoff, refresh tokens

9.3 Infrastructure Issues

Symptom	Possible Cause	Diagnostic Command	Resolution
Kafka broker down	Disk failure, JVM OOM	<code>docker logs kafka-broker-1</code>	Restart broker, check disk/JVM heap
ES cluster yellow/red	Shard allocation failed, node missing	<code>curl cluster/health?pretty</code>	Fix allocation, recover node
Grafana datasource error	ES/Kafka credentials expired	Test connection in Grafana UI	Rotate credentials, update datasources
Network partition	Switch failure, firewall rule change	ping, traceroute to cluster nodes	Engage network ops, verify firewall

Symptom	Possible Cause	Diagnostic Command	Resolution
Clock skew between nodes	NTP failure, VM drift	ntpq -p on all nodes	Fix NTP config, sync clocks

9.4 Performance Tuning Recommendations

When encountering performance bottlenecks, apply these tuning adjustments in order, measuring impact after each change. Avoid changing multiple parameters simultaneously as this obscures which adjustment produced observed effects. Document all tuning changes in the operations runbook for future reference and rollback capability.

Component	Parameter	Default	Tuned Value	Effect
Kafka	socket.receive.buffer.bytes	65536	262144	Higher throughput
Kafka	num.io.threads	8	16	Better disk I/O utilization
Elasticsearch	index.refresh_interval	1s	5s	Reduced indexing overhead (+400% throughput)
Collector	pcap.snaplen	65535	4096	Smaller captures (if headers sufficient)
Analyzer	rule_evaluation_threads	CPU count	CPU count x 2	Parallel rule execution
JVM (all Java services)	mx	1g	4g	More heap for caching
Linux kernel	net.core.somaxconn	128	1024	Higher connection backlog

Performance Testing Protocol

Before deploying to production, conduct load testing using captured PCAP replays at 1.5x expected peak traffic. Measure end-to-end latency (capture to alert), resource utilization (CPU, memory, disk I/O), and Kafka consumer lag under sustained load. Document baseline metrics and establish alert thresholds at 70% of maximum tested capacity to provide headroom for traffic spikes.

Appendix A: Port Reference

Port	Protocol	Service	Direction	Security Zone
2904	M3UA	SS7 over SIGTRAN	Inbound (passive)	soc-events
2905	SCTP	Transport layer	Inbound (passive)	soc-events
3868	Diameter	LTE/EPS signaling	Inbound (passive)	soc-events
3869	Diameter (alt)	Diameter Monitor mgmt	Inbound	soc-events
7000	HTTP	SS7 Collector health	Internal	soc-backend
7001	HTTP	SS7 Analyzer API	Internal	soc-backend
9092	PLAINTEXT	Kafka broker internal	Internal	soc-events
9200	HTTP	Elasticsearch REST	Internal	soc-backend
9000	HTTP	TheHive web UI	Internal	soc-backend
3000	HTTP	Grafana dashboards	Internal	soc-monitoring

Appendix B: Glossary

Term	Definition
Diameter	IP-based protocol suite for AAA (Authentication, Authorization, Accounting) in LTE/IMS networks, successor to RADIUS. Used for authentication and authorization.
Global Title (GT)	Address used in SS7 networks to route messages to destination, similar to phone numbers but for signaling. Format varies by network.
HLR (Home Location Register)	Database containing subscriber profile, current location (MSC/VLR address), and service entitlements for home network subscribers.
IMSI (International Mobile Subscriber Identity)	Unique subscriber identifier stored on SIM card, typically 15 digits. Used for network authentication and subscriber identification.
IRSF (International Revenue Share Fraud)	Attack where attackers generate high-volume international calls to premium-rate numbers they control, sharing revenue with legitimate carriers.
ISUP (ISDN User Part)	SS7 protocol layer responsible for setting up and tearing down telephone calls, part of MTP3 user layer alongside MAP and CAP.
MAP (Mobile Application Part)	SS7 application protocol for GSM/UMTS mobility management including location updates, authentication, SMS delivery, and call control.
M3UA (MTP3 User Adaptation Layer)	SIGTRAN adaptation layer that transports MTP3 messages over IP using SCTP instead of traditional TDM links.
MSC (Mobile Switching Center)	Core network element that handles call switching, mobility management, and interfaces between radio access network and core network.
PCAP (Packet Capture)	File format for storing captured network packets, used for forensic analysis and replay testing. Tools: tcpdump, Wireshark.

Term	Definition
SIGTRAN	IETF working group and protocol suite for transporting SS7 messages over IP networks, includes M3UA, SUA, M2PA, and S
SCTP (Stream Control Transmission Protocol)	Transport layer protocol designed for telecommunications signaling, provides reliable ordered delivery with multi-streaming a
STP (Signal Transfer Point)	Network node that routes SS7 messages between signaling endpoints (like IP routers but for SS7), critical infrastructure for s
USSD (Unstructured Supplementary Service Data)	Interactive services (balance inquiry, recharge) initiated by dialing service codes like *100#.
VLR (Visitor Location Register)	Temporary database storing profiles of subscribers currently roaming in an MSC service area, synchronized with HLR for loc

Appendix C: Version History

Version	Date	Author	Changes
1.0	2025-08-01	SOC Engineering Team	Initial release - architecture, services, rules
1.1	2025-08-10	SOC Engineering Team	Added Diameter monitoring, LTE interfaces
2.0	2026-08-03	Documentation Team	Complete rewrite - operations, compliance, troubleshooting